

Relational Database Design

Join Dependencies in Practice

Uni Auckland Databases 51

Gerald Weber

Recap: Inferring decompositions from Data

- ▶ We frequently infer heuristically such constraints from database instances and the patterns we can see.
- ▶ e.g. **Course(course_id, hod_name, dept_name, credits)**.

▶ Tuples:

course_id	hod_name	dept_name	credits
C1	Pat	CS	4
C2	Pat	CS	3
C3	Kim	Math	3

- ▶ We assume the repeated name Pat is not a coincidence.
- ▶ Once we made this observation, we assume only relations are legal where this pattern is maintained.
 - ▶ **Course1(course_id, dept_name)**
 - ▶ **Course2(dept_name, hod_name)**
 - ▶ **Course3(course_id, credits)**

Example Dining Option Table

- ▶ Consider a table representing restaurants, their menus, and desserts.
- ▶ **Restaurant(restaurant_name, menu_name, dessert_name).**

▶ Tuples:

restaurant_name	menu_name	dessert_name
Bistro	Budget	Fruits
Bistro	Budget	Berry
Bistro	Express	Fruits
Bistro	Express	Berry
Diner	Gourmet	Treats
Diner	Gourmet	Berry
Diner	Deluxe	Treats
Diner	Deluxe	Berry

- ▶ Is this a good Design? Do we need to decompose it?

Example Dining Option Table

- ▶ **Restaurant(restaurant_name, menu_name, dessert_name).**

- ▶ Tuples:

restaurant_name	menu_name	dessert_name
Bistro	Budget	Fruits
Bistro	Budget	Berry
Bistro	Express	Fruits
Bistro	Express	Berry
Diner	Gourmet	Treats
Diner	Gourmet	Berry
Diner	Deluxe	Treats
Diner	Deluxe	Berry

- ▶ We observe that each restaurant has multiple menus and multiple desserts. **Normalization:** Decomposition:
 - ▶ **Restaurant2(restaurant_name, menu_name)**
 - ▶ **Restaurant3(restaurant_name, dessert_name)**

Decomposed Relations

► Restaurant(restaurant_name, menu_name, dessert_name)

restaurant_name	menu_name	dessert_name
Bistro	Budget	Fruits
Bistro	Budget	Berry
Bistro	Express	Fruits
Bistro	Express	Berry
Diner	Gourmet	Treats
Diner	Gourmet	Berry
Diner	Deluxe	Treats
Diner	Deluxe	Berry

► Decomposed Relations:

Restaurant2

r..._name	menu_name
Bistro	Budget
Bistro	Express
Diner	Gourmet
Diner	Deluxe

Restaurant3

r..._name	dessert_name
Bistro	Fruits
Bistro	Berry
Diner	Treats
Diner	Berry

Example UPI

- ▶ Assume this table in our Uni
- ▶ e.g. **User(ID, Givenn, Surn, G_UPI, Sur_UPI, num_UPI), degree.**

- ▶ Tuples:

ID	Givenn	Surn	Gv_UPI	Sur_UPI	num_UPI	year
1	John	Smith	j	smi	123	2024
2	Jane	Doe	j	doe	456	2023
3	Robert	Brown	r	bro	789	2023

- ▶ The attributes Given_UPI and Sur_UPI are derived from Givenname and Surname, respectively.
- ▶ The num_UPI attribute ensures the uniqueness of the UPI.
- ▶ Should we decompose this table?

Recap: Aim

- ▶ If a relation schema R can be reconstructed just by natural join from a set of schemata $R_1, R_2, \dots, R_n$ with all of them subsets of R :
- ▶ then we prefer a database with relations based on $R_1, R_2, \dots, R_n$
- ▶ we have to identify what concerns we have to take into account in this process.

Recap: Formalization

- ▶ Note $\text{Course}(\text{course_id}, \text{hod_name}, \text{dept_name}, \text{credits})$ both defines Course and states that it has the schema $\{\text{course_id}, \text{hod_name}, \text{dept_name}, \text{credits}\}$.
- ▶ Given relation schema R and a set of schemata $R_1, R_2, \dots, R_n$:
- ▶ we want to formalize the statement:
- ▶ R can be reconstructed just by natural join from $R_1, R_2, \dots, R_n$. iff for all legal $r(R)$:
$$r = \Pi_{R_1}(r) \bowtie \Pi_{R_2}(r) \bowtie \dots \bowtie \Pi_{R_n}(r)$$
- ▶ Different wording: R can be reconstructed by joining multiple projections of itself.
- ▶ We call this a Join Dependency.
- ▶ Syntax 1 : **JD($R, [R_1, R_2, \dots, R_n]$)**
- ▶ Syntax 2: $\ast(R_1, R_2, \dots, R_n)$ on schema R

Restate Aim

- ▶ If a relation schema R can be reconstructed just by natural join from a set of schemata $R_1, R_2, \dots, R_n$ with all of them subsets of R :
- ▶ then we observe a Join Dependency: $JD(R, [R_1, R_2, \dots, R_n])$
- ▶ then we prefer decomposition of the table, a database with relations based on $R_1, R_2, \dots, R_n$
- ▶ Note, this at first leads to a separate table for every attribute dependent on a key.
 - ▶ This is akin to a column store database.
 - ▶ This is a valid and sometimes preferred design.
 - ▶ However, tables with the same key can then be merged (joined) on demand.

Recap: Decomposition (Column Store Style)

- ▶ Decomposition of a schema R into schemas $R_1, R_2, \dots, R_n$.
- ▶ Consider the relation **Course(course_id, title, dept_name, credits)**.

- ▶ Tuples:

course_id	title	dept_name	credits
C1	Database Systems	CS	4
C2	Algorithms	CS	3
C3	Linear Algebra	Math	3

- ▶ The relation can be reconstructed by joining:
 - ▶ **Course1(course_id, title)**
 - ▶ **Course2(course_id, dept_name)**
 - ▶ **Course3(course_id, credits)**
- ▶ on **course_id**.